



RNDr. Šimon Mackovjak, PhD.
Department of Space Physics, Institute of Experimental Physics,
Slovak Academy of Sciences, Watsonova 47, 040 01 Košice, Slovakia
mackovjak@saske.sk



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Dear Selection Committee,

I am pleased to provide a strong recommendation for Lukáš Randuška in support of his application for financial assistance to attend the European Space Weather Week 2026. I have been supervising Lukáš at the Institute of Experimental Physics of the Slovak Academy of Sciences in Košice, where he has been working as an intern alongside his Bachelor's studies in Data Science and Artificial Intelligence at Pavol Jozef Šafárik University.

Lukáš has spent the past three years developing a comprehensive CME detection-to-impact pipeline for geomagnetic storm prediction — a task that would challenge many postgraduate researchers, let alone an undergraduate. His pipeline begins with a CACTus-inspired CME detection algorithm applied to LASCO C2 coronagraph images, which he validated to perform on par with the original CACTus method, achieving a velocity mean absolute error of 160 km/s. Crucially, Lukáš went beyond detection: he coupled this front end to a physics-informed neural network (PINN), pre-trained on synthetic events generated as solutions of the extended drag-based model, to predict CME transit time from parameters observable at onset. The predicted transit time is then used to reconstruct a 1D velocity time-series of the CME's journey from Sun to Earth, providing upstream input for a Dst/SYM-H prediction model. This end-to-end approach directly addresses the central challenge of space weather forecasting: generating accurate geomagnetic storm warnings not just minutes, but potentially more than a day, before impact.

What distinguishes Lukáš is not only the technical depth of his work, but the independence and initiative with which he has pursued it. Over three years, he has built up a serious understanding of solar wind physics, machine learning, and operational forecasting — a rare combination at the undergraduate level. His contribution aligns squarely with the ESWW 2026 theme of combining data-driven and physics-based, cross-scale forecasting, and the conference represents an excellent opportunity for him to present his results to an international audience, receive constructive feedback from leading experts in the field, and identify potential collaborators. This would be his first international conference, and I believe attending it would be a formative step in his development as a researcher.

I wholeheartedly support Lukáš's application for financial support and recommend him without reservation. Should you require any further information, please do not hesitate to contact me.

Yours sincerely,

RNDr. Šimon Mackovjak, PhD.